

COMS BC1016

Introduction to Computational Thinking and Data Science

Lecture 7: Groups and Pivots

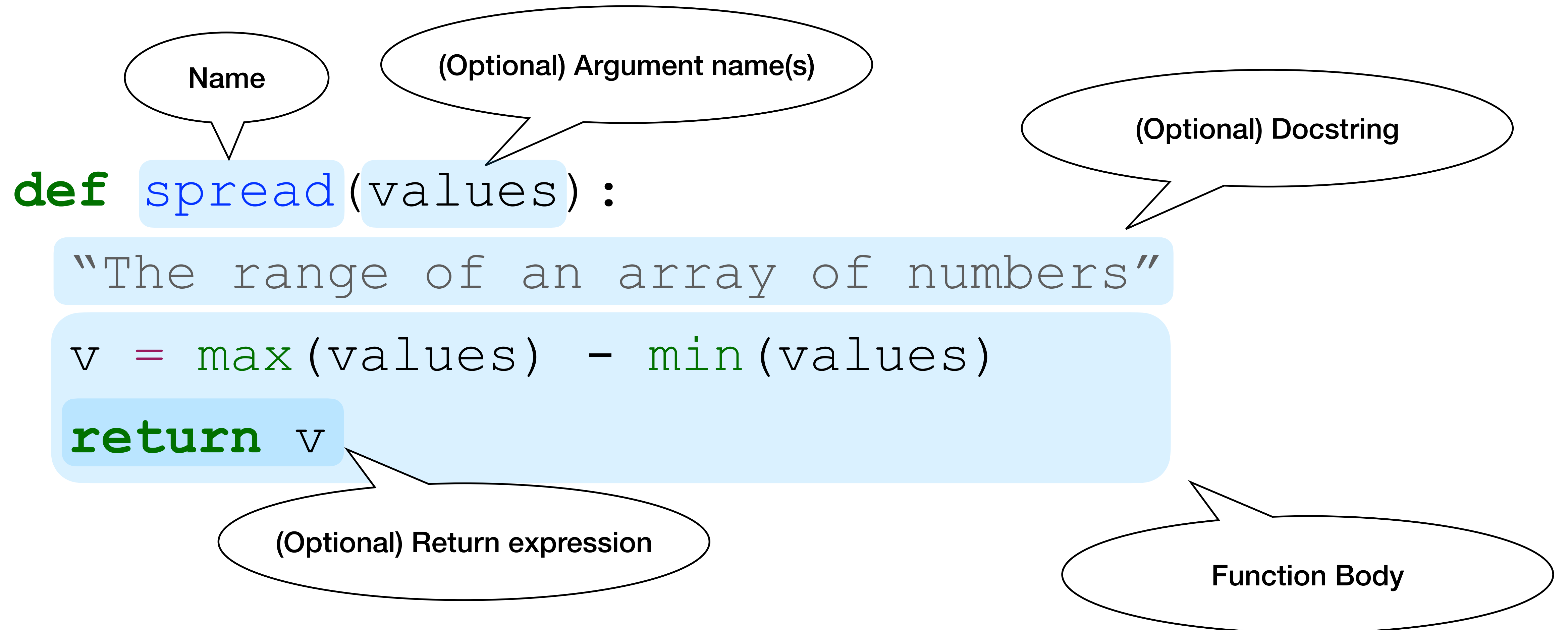
BARNARD COLLEGE OF COLUMBIA UNIVERSITY

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Functions (Continued)

Last Time: Anatomy of a Function

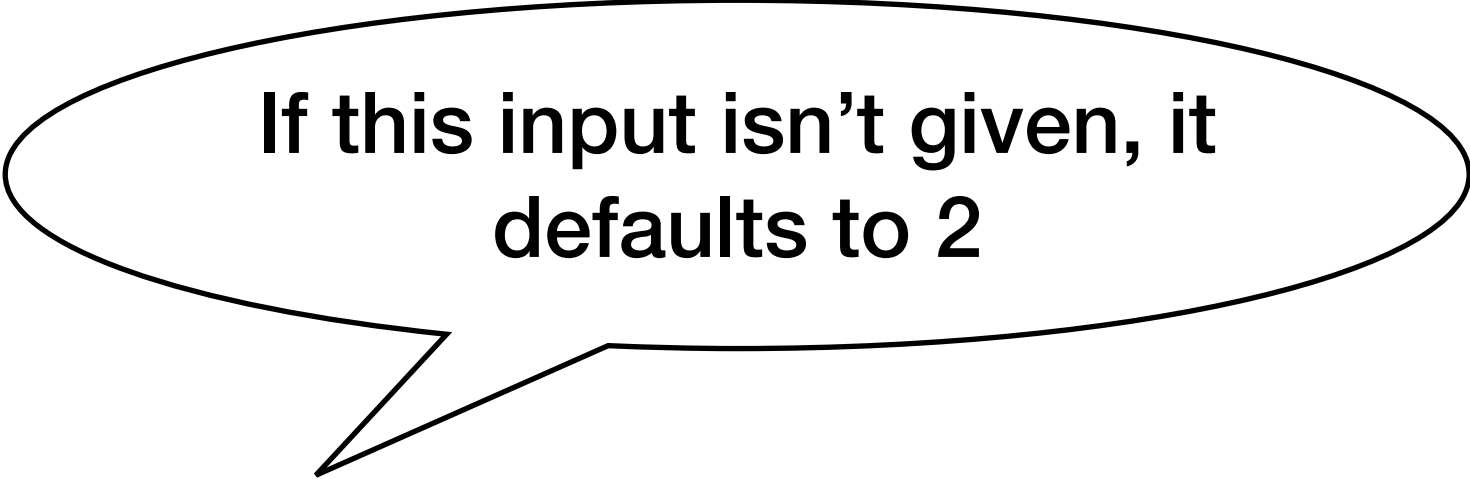
Use **def** to define your own function!



Demo: Function Practice

Default Values for Function Inputs

You can specify a default value for a function input to make the input optional!



If this input isn't given, it defaults to 2

```
def rounded_frac(x, y, decimal_place=2):  
    "Divides x by y and rounds the decimal"  
    v = (x/y).round(decimal_places)  
return v
```

Manipulating Table Data with `apply`

`apply`

1. Calls a function on each element in the input column(s)
2. Produces an array containing the output of the function on each input column element

```
tbl.apply(function_name, column_label)
```

```
tbl.apply(function_name, column_label1, column_label2)
```

DEMO

Grouping by One Column

Grouping by One Column

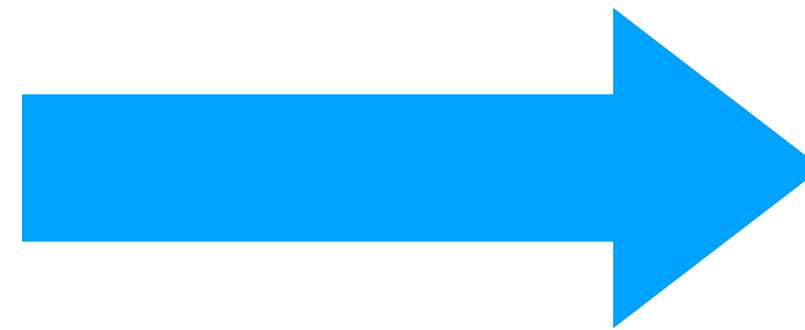
group aggregates all rows with the same value for a column into a single row in a resulting table

- First argument: Which column to group by
- Second argument: (Optional) How to combine values
 - **len**: (default) number of grouped values
 - **list**: List of all grouped values
 - **sum**: total of all grouped values

Simple Grouping

Name	Age	Weight	Coloring
Ruby	14	8	tuxedo
Gertrude	15	12	tuxedo
Hamby	8	16	tabby
Fig	3	7	tabby
Corina	6	10	tortie
Frito	2	8.5	tabby

```
cat_tbl.group('Coloring')
```



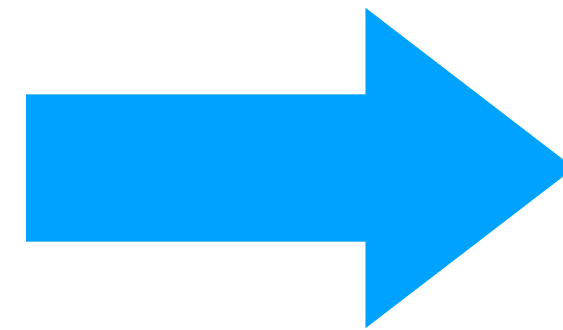
Coloring	count
tabby	3
tuxedo	2
tortie	1

Counts and other columns are ignored.

Grouping with a Function

Name	Age	Weight	Coloring
Ruby	14	8	tuxedo
Gertrude	15	12	tuxedo
Hamby	8	16	tabby
Fig	3	7	tabby
Corina	6	10	tortie
Frito	2	8.5	tabby

```
cat_tbl.group('Coloring', np.average)
```



Coloring	Name average	Age average	Weight average
tabby		4.33333	10.5
tortie		6	10
tuxedo		14.5	10

All other columns are averaged.

Lists and Grouping by Multiple Columns

Lists are Generic Sequences

A **list** in Python is a sequence of values (just like an array) **but** the values can have different types:

```
[2+3, 'four', 5.0]
```

Lists and Groups

- Grouping by a list makes one row per *combination* of unique values
- This kind of grouping is useful for *cross-classification*: counting combinations of a pair of attributes

Name	Age	Weight	Coloring	Sex
Ruby	14	8	tuxedo	F
Gertrude	15	12	tuxedo	F
Hamby	8	16	tabby	M
Fig	3	7	tabby	F
Corina	6	10	tortie	F
Frito	2	8.5	tabby	M

```
cat_tbl.group(['Coloring', 'Sex'])
```



Coloring	Sex	count
tabby	F	1
tabby	M	2
tortie	F	1
tuxedo	F	2

Pivot Tables

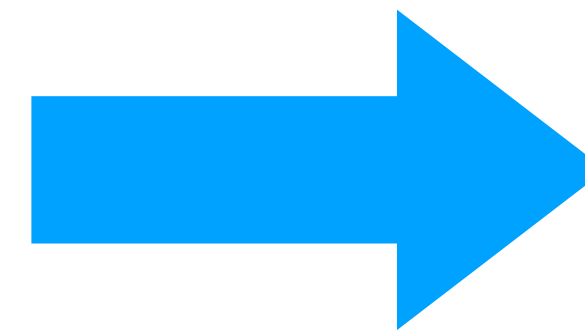
Pivot Tables

- Cross-classifies according to two categorical variables
- Produces a grid of counts or aggregated values
- Two required arguments: `tbl.pivot(col_var, row_var)`
- Two optional arguments (include **both** or **neither**):
`tbl.pivot(col_var, row_var, values, collect)`
 - `values`: column label to aggregate
 - `collect`: function to aggregate with

Pivot Tables

```
cat_tbl.pivot('Owner', 'Sex')
```

Name	Age	Weight	Coloring	Sex	Owner
Ruby	14	8	tuxedo	F	Alice
Gertrude	15	12	tuxedo	F	Alice
Hamby	8	16	tabby	M	Bob
Fig	3	7	tabby	F	Bob
Corina	6	10	tortie	F	Carol
Frito	2	8.5	tabby	M	Carol

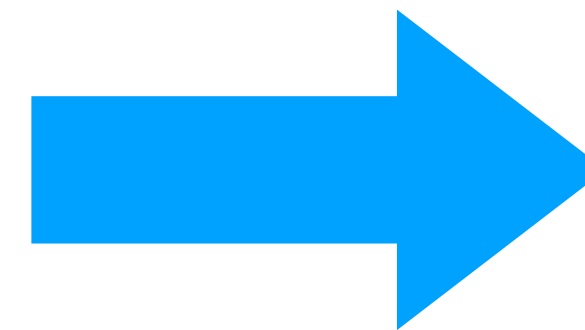


Sex	Alice	Bob	Carol
F	2	1	1
M	0	1	1

Pivot Tables

```
cat_tbl.pivot('Owner', 'Sex', 'Age', np.average)
```

Name	Age	Weight	Coloring	Sex	Owner
Ruby	14	8	tuxedo	F	Alice
Gertrude	15	12	tuxedo	F	Alice
Hamby	8	16	tabby	M	Bob
Fig	3	7	tabby	F	Bob
Corina	6	10	tortie	F	Carol
Frito	2	8.5	tabby	M	Carol



Sex	Alice	Bob	Carol
F	14.5	3	6
M	0	8	2

Group vs Pivot

Group

- One combo of grouping variables **per row**
- **Any number** of grouping variables
- Aggregate values of **all other columns** in the table
- Missing combos are **absent**

```
cat_tbl.group(['Sex', 'Coloring'], np.average)
```

Sex	Coloring	Name average	Age average	Weight average	Owner average
F	tabby		3	7	
F	tortie		6	10	
F	tuxedo		14.5	10	
M	tabby		5	12.25	

Pivot

- One combo of grouping variables **per entry**
- **Two** grouping variables: columns and rows
- Aggregate values of **values column**
- Missing combos = **0 (or empty string)**

```
cat_tbl.pivot('Sex', 'Coloring', 'Weight', np.average)
```

Coloring	F	M
tabby	7	12.25
tortie	10	0
tuxedo	10	0

Tips for Programming

Solving Table Manipulation Problems

1. **Understand the result:** What information is needed to fill in a particular value of the goal?
2. **Describe the operations:** articulate (in English) the operations that will produce the result
3. **Write Python expressions:** express the operations using table methods, functions, and array arithmetic

If you're stuck...

- If you don't know where to start, think how you would do it manually
 - What would you do if you were solving on pen and paper?
 - Optionally, solve a simplified version (e.g., less data or smaller values).
 - Once you get the simpler version working, expand to the original version
- If you have code, break it into smaller pieces and test each part separately
 - What does each component do? Is it what you expect/want?
 - What data type is being produced? (You can check using `type( . . )`)

Next Class

- Today
 - Groups
 - Lists
 - Pivots
- Monday
 - Conditionals